

ECON 165: International Finance

Section: Week 7

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Exchange rate fluctuations in the short-run

- Assumptions:
 - Prices are sticky: $\bar{P}, \bar{P}^*$
 - Government and taxes are fixed: $\bar{G}, \bar{T}$
 - Foreign output and interest rates are fixed: $\bar{Y}^*, \bar{i}^*$
 - Exchange rate expectations are fixed: $\bar{E}^e$
- Consumption is increasing in disposable income: $C(Y - \bar{T})$
- Investment is decreasing in the interest rate: $I(i)$
- Trade balance is increasing in the real exchange rate, decreasing in home disposable income, and increasing in ROW disposable income:

$$TB \left(\frac{E\bar{P}^*}{\bar{P}}, Y - \bar{T}, \bar{Y}^* - \bar{T}^* \right)$$

Goods market + FX market equilibria $\longrightarrow$ IS curve

IS curve: combinations of output Y and the interest rate i for which the goods and FX markets are in equilibrium

- Goods market equilibrium in a *closed* economy

$$Y = C(Y - \bar{T}) + I(i) + \bar{G}$$

Goods market + FX market equilibria $\longrightarrow$ IS curve

IS curve: combinations of output Y and the interest rate i for which the goods and FX markets are in equilibrium

- Goods market equilibrium in an *open* economy

$$Y = C(Y - \bar{T}) + I(i) + \bar{G} + TB \left(E \frac{\bar{P}^*}{\bar{P}}, Y - \bar{T}, \bar{Y}^* - \bar{T}^* \right)$$

Goods market + FX market equilibria $\longrightarrow$ IS curve

IS curve: combinations of output Y and the interest rate i for which the goods and FX markets are in equilibrium

- Goods market equilibrium

$$Y = C(Y - \bar{T}) + I(i) + \bar{G} + TB\left(E \frac{\bar{P}^*}{\bar{P}}, Y - \bar{T}, \bar{Y}^* - \bar{T}^*\right)$$

- FX market equilibrium

$$i = \bar{i}^* + \left(\frac{\bar{E}^e}{E} - 1\right) \quad \Longrightarrow \quad E = \frac{\bar{E}^e}{i - \bar{i}^* + 1}$$

Goods market + FX market equilibria $\longrightarrow$ IS curve

IS curve: combinations of output Y and the interest rate i for which the goods and FX markets are in equilibrium

- Goods market equilibrium

$$Y = C(Y - \bar{T}) + I(i) + \bar{G} + TB\left(\frac{\bar{E}^e}{i - \bar{i}^* + 1} \frac{\bar{P}^*}{\bar{P}}, Y - \bar{T}, \bar{Y}^* - \bar{T}^*\right)$$

- FX market equilibrium

$$i = \bar{i}^* + \left(\frac{\bar{E}^e}{E} - 1\right) \implies E = \frac{\bar{E}^e}{i - \bar{i}^* + 1}$$

Example: Deriving the IS curve

Suppose goods market equilibrium is

$$Y = \underbrace{0.8(Y - \bar{T})}_C + \underbrace{10 - 15i}_I + \bar{G} + \underbrace{20 \left(1 - \frac{1}{E}\right) - 0.3(Y - \bar{T})}_{TB}$$

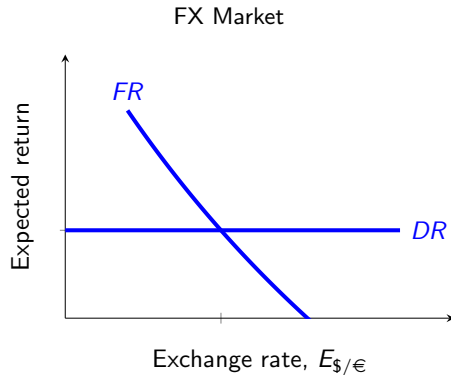
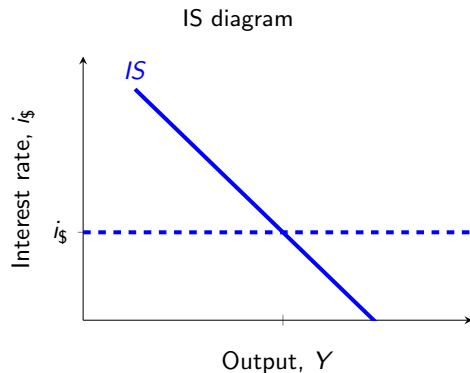
and

- $\bar{G} = 4.2$
- $\bar{T} = 10$
- $\bar{i}^* = 0.1$
- $\bar{E}^e = 1$

Simplifying, we can write the IS curve as

$$i = 0.32 - \frac{Y}{70}$$

Example: Deriving the IS curve



★ At every point on the IS curve, the goods market and FX market are in equilibrium.

Example: Shifts in the IS curve

Suppose government expenditure **increases** to $\bar{G} = 6.2$:

$$Y = \underbrace{0.8(Y - \bar{T})}_C + \underbrace{10 - 15i}_I + \bar{G} + \underbrace{20\left(1 - \frac{1}{E}\right) - 0.3(Y - \bar{T})}_{TB}$$

The new IS curve is

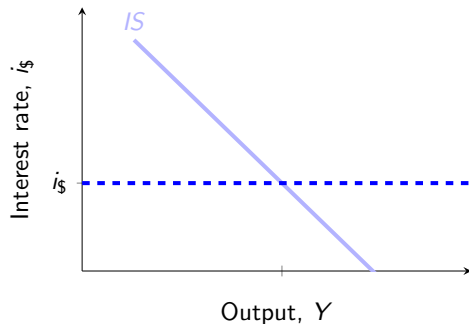
$$i = 0.377 - \frac{Y}{70}$$

(compared with the old IS curve $i = 0.32 - \frac{Y}{70}$).

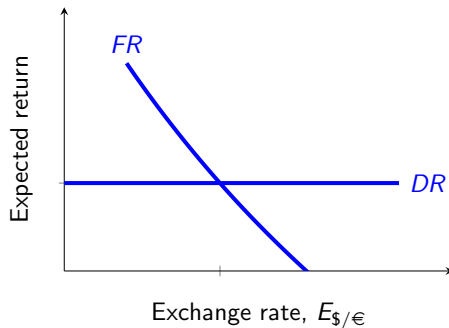
Example: Shifts in the IS curve

What does the new IS curve look like?

IS diagram

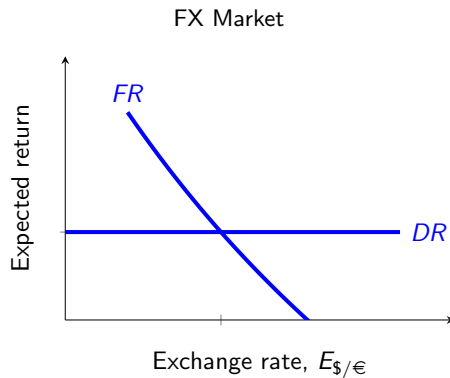
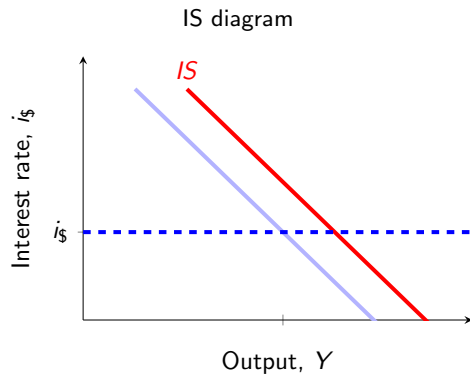


FX Market



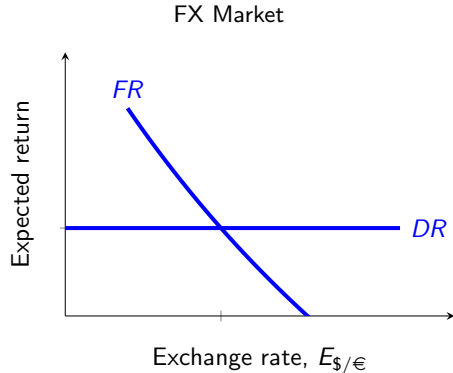
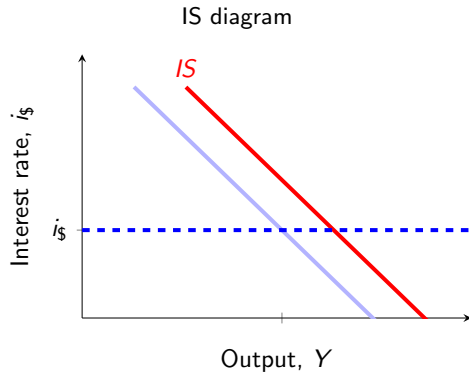
Example: Shifts in the IS curve

What does the new IS curve look like?



Example: Shifts in the IS curve

What does the new IS curve look like?



What else causes the IS curve to **shift**? What causes us to **move along** the IS curve?

Money market equilibrium $\longrightarrow$ LM curve

LM curve: combinations of output Y and the interest rate i for which the money market is in equilibrium

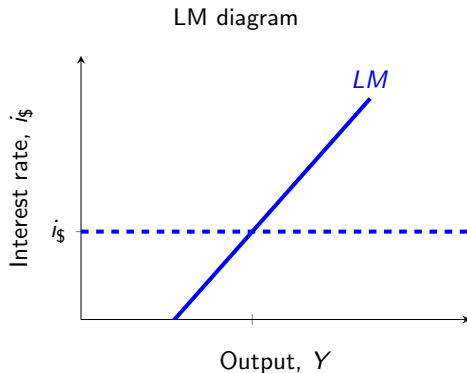
- Money market equilibrium

$$\frac{\overline{M}}{\overline{P}} = L(i, Y)$$

Example: Deriving the LM curve

Suppose the LM curve is $\frac{\bar{M}}{\bar{P}} = Y - 70i$, with $\bar{M} = 8.4$ and $\bar{P} = 1$. We can rewrite this as

$$8.4 = Y - 70i \quad \Rightarrow \quad i = \frac{Y}{70} - 0.12.$$



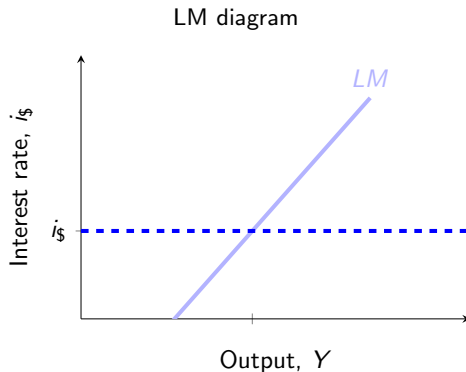
★ At every point on the LM curve, the money market is in equilibrium.

Example: Shifts in the LM curve

Suppose the money supply increases to $\bar{M} = 15$. The new LM curve is

$$15 = Y - 70i \quad \Rightarrow \quad i = \frac{Y}{70} - 0.214.$$

What does the new LM curve look like?

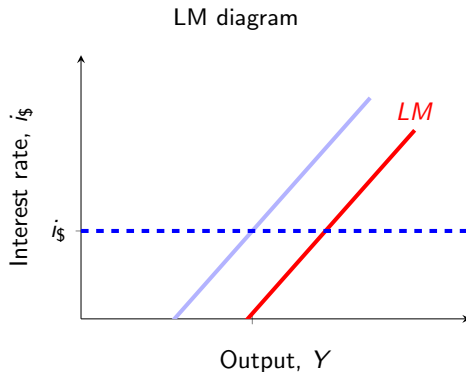


Example: Shifts in the LM curve

Suppose the money supply increases to $\bar{M} = 15$. The new LM curve is

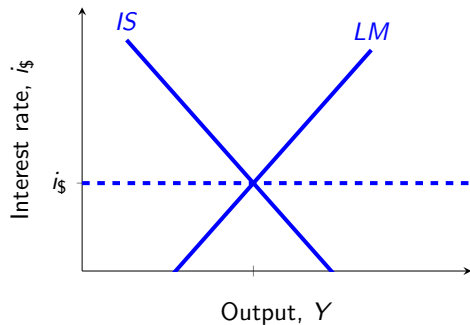
$$15 = Y - 70i \quad \Rightarrow \quad i = \frac{Y}{70} - 0.214.$$

What does the new LM curve look like?

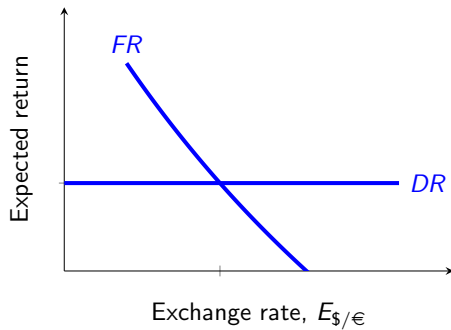


IS-LM-FX diagram: All markets in equilibrium

IS-LM diagram

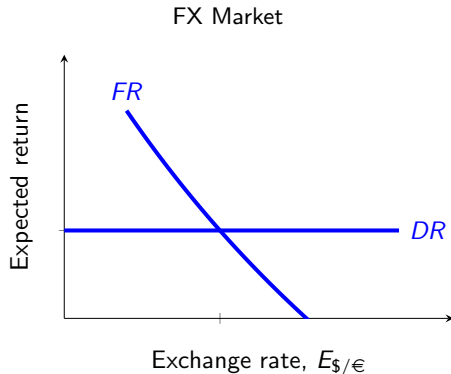
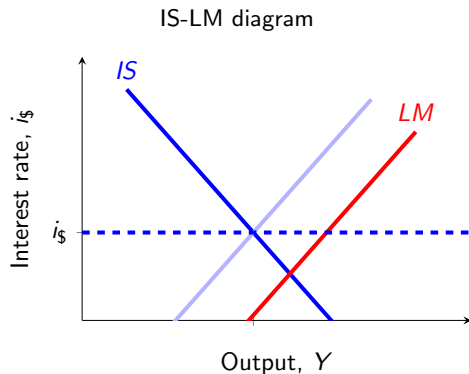


FX Market



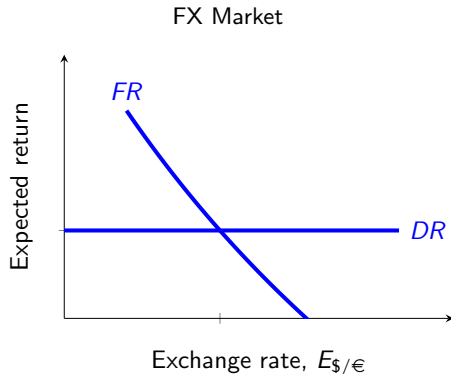
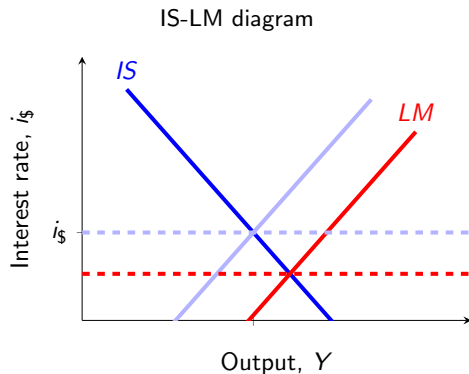
Application: Monetary policy under a floating exchange rate regime

Suppose the money supply increases, shifting the LM curve to the right. How do markets adjust to reach equilibrium?



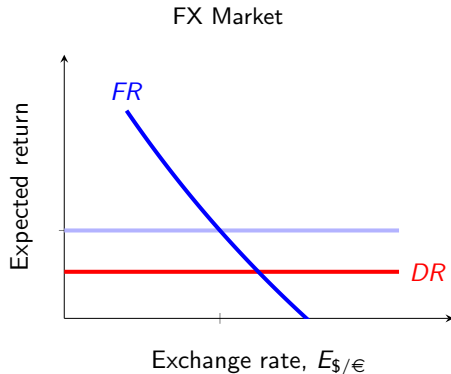
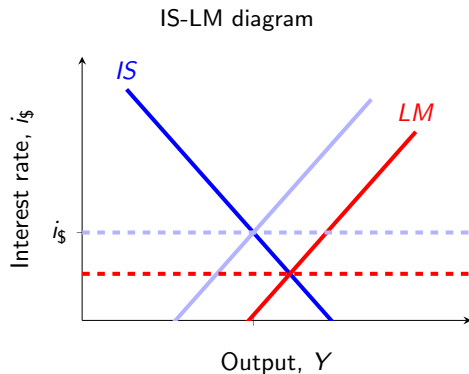
Application: Monetary policy under a floating exchange rate regime

Suppose the money supply increases, shifting the LM curve to the right. Equilibrium in all markets requires the domestic interest rate to fall...



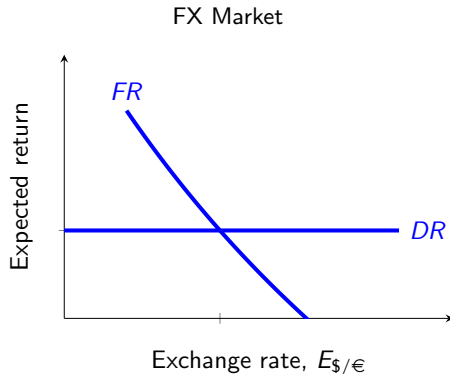
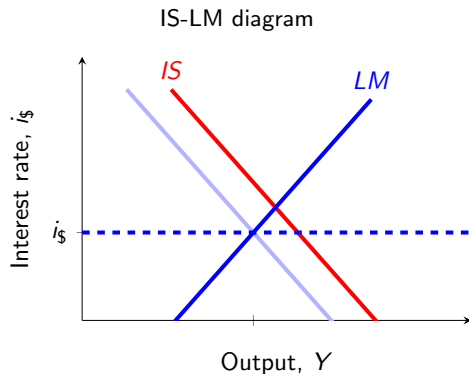
Application: Monetary policy under a floating exchange rate regime

Suppose the money supply increases, shifting the LM curve to the right. Equilibrium in all markets requires the domestic interest rate to fall, causing the exchange rate to **depreciate**.



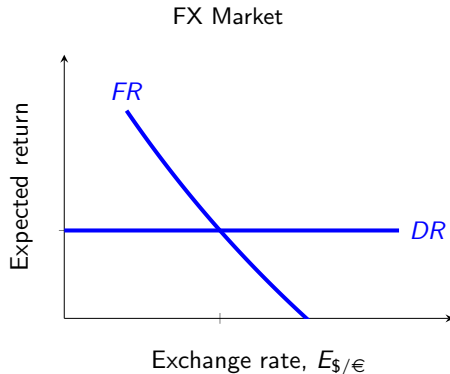
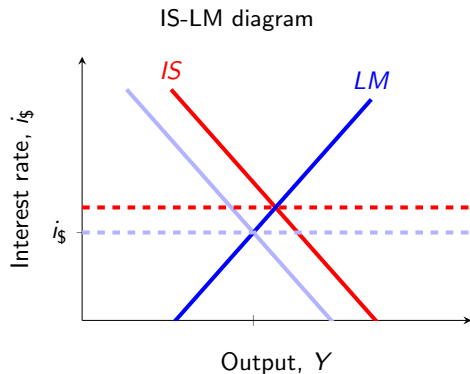
Application: Fiscal policy under a floating exchange rate regime

Suppose the government expenditure increases, shifting the IS curve to the right. How do markets adjust to reach equilibrium?



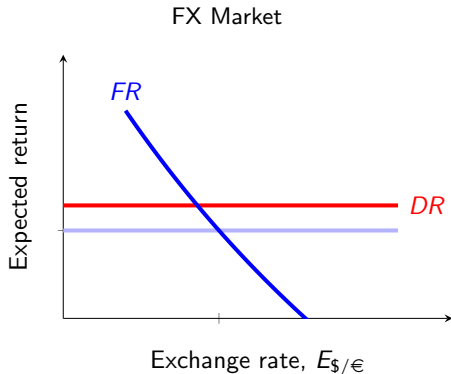
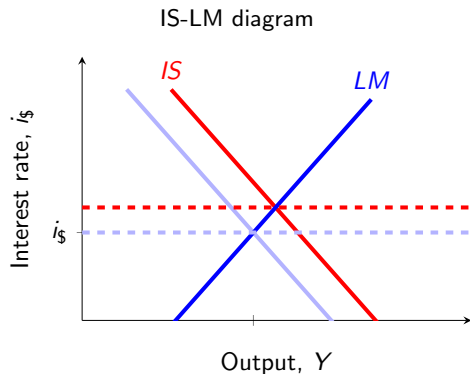
Application: Fiscal policy under a floating exchange rate regime

Suppose the government expenditure increases, shifting the IS curve to the right. Equilibrium in all markets requires the domestic interest rate to rise...



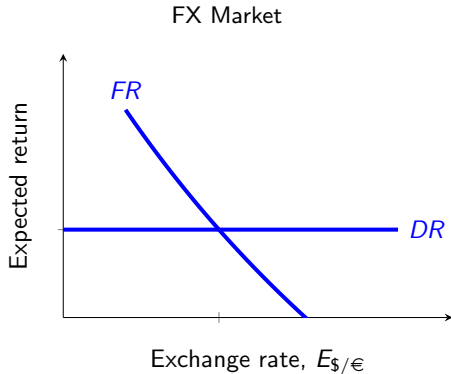
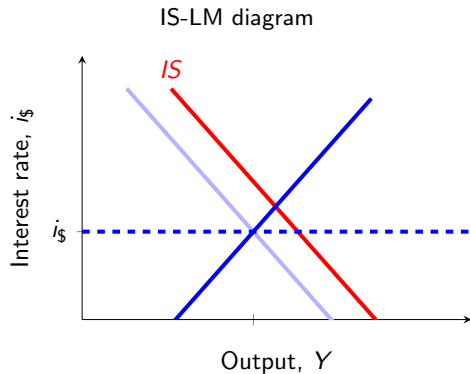
Application: Fiscal policy under a floating exchange rate regime

Suppose the government expenditure increases, shifting the IS curve to the right. Equilibrium in all markets requires the domestic interest rate to rise, causing the exchange rate to **appreciate**.



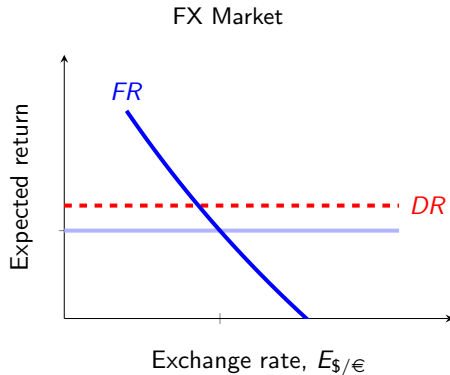
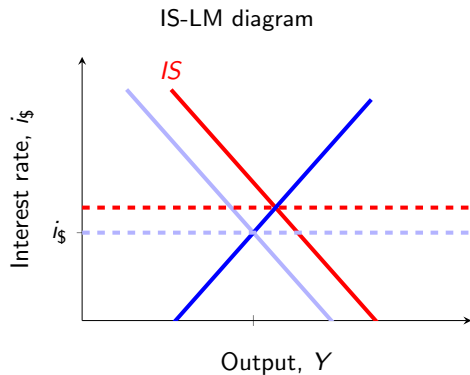
Application: Fiscal policy under a fixed exchange rate regime

Suppose the government expenditure increases, shifting the IS curve to the right.



Application: Fiscal policy under a fixed exchange rate regime

Suppose the government expenditure increases, shifting the IS curve to the right. As we just saw, equilibrium in all markets requires the domestic interest rate to rise, causing the exchange rate to appreciate.



Application: Fiscal policy under a fixed exchange rate regime

Suppose the government expenditure increases, shifting the IS curve to the right. As we just saw, equilibrium in all markets requires the domestic interest rate to rise, causing the exchange rate to appreciate. To counteract this appreciation, money supply must increase.

